

The Pair-Creation Threshold of Impulsive Stueckelberg–Horwitz–Piron Scattering Is a Cayley-Transform Pole

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In classical Stueckelberg electrodynamics a particle worldline represents pair annihilation when its time coordinate reverses direction with respect to the evolution parameter. A recent calculation by Land exhibits this reversal in impulsive Coulomb scattering and finds a sharp threshold, the reversal occurring exactly when a dimensionless interaction strength exceeds two, equivalently when the interaction energy exceeds the rest energy of the annihilated pair. We replicate the published calculation exactly and then locate the origin of its threshold. The impulsive midpoint convention used to close the scattering equations is, for the reduced two-dimensional kick system, precisely the Cayley transform of the interaction generator, which is the lowest diagonal Padé approximant of the exponential map. The published threshold is the pole of that approximant. Integrating the same generator continuously instead gives hyperbolic evolution that conserves a quadratic form and provably never reverses the time velocity from physical initial data, at any interaction strength. The conserved form is bounded below on physical data, which sharpens that statement to the time velocity never falling below its rest value. Because the reduced generator is a constant matrix, the continuously accumulated map is moreover the exact solution for every smooth unit-mass kernel at every width, so it is the common limit of all smooth regularizations of the impulse and the midpoint map beyond the pole is the limit of none. In group language, the continuous flow stays in the identity component of the invariance group of the conserved form, while beyond the pole the Cayley transform lands in the disconnected component containing the simultaneous reversal of time velocity and radial velocity, which no continuous flow of the generator can reach. The published algebra is correct as published. What our result shows is that its classical pair event marks the breakdown radius of an approximant rather than a dynamical transition of the continuously accumulated force, and that whether the full field dynamics restores a genuine threshold is an open question. A practical corollary for simulation work is that reversal statistics can be created by a discretization prescription alone.

I. INTRODUCTION

Stueckelberg proposed that a single classical worldline can describe pair annihilation if its laboratory time coordinate, viewed as a function of an independent evolution parameter, turns around [1, 2]. A worldline that turns carries one segment running backward in laboratory time, read as the antiparticle in Feynman’s treatment of the same picture [8], and the vertex at the turn is a creation or an annihilation according to the direction in which the line is traversed. Both names are used in the literature for the one event, the source replicated here calls it production [4], and this paper uses the terms interchangeably for the single reversal. No distinction is intended anywhere below. The framework of Horwitz and Piron gives this idea a covariant Hamiltonian mechanics [3], and its gauge-invariant electrodynamics permits the mass exchange with the field that a turnaround requires [5]. Within that theory, Land computed a concrete example [4]. An event scatters impulsively in the Coulomb field of a heavy nucleus, and its outgoing time velocity reverses sign exactly when a dimensionless coupling, built from the correlation length, the interaction distance, and the charges, exceeds two. The condition translates to the

statement that the interaction energy exceeds the rest energy of the annihilated pair, which is physically appealing, and the calculation is fully explicit.

This paper reports a replication of that calculation and one structural finding about it. The replication is exact. Every printed equation we transcribe passes machine-precision consistency guards, the threshold appears exactly where the source places it, and the limiting outgoing energy agrees with the printed asymptote. The finding concerns where the threshold comes from. The source closes its impulsive scattering equations with a midpoint convention, evaluating the interaction with the average of the incoming and outgoing velocities. For the reduced system that governs the time velocity and the radial velocity component, that convention is exactly the Cayley transform of the interaction generator. The Cayley transform is the simplest diagonal Padé approximant of the exponential map, and it has a pole. The published threshold sits exactly on that pole. The alternative closure, continuous accumulation of the same generator, has no pole. It conserves a quadratic form that confines the motion to one branch of a hyperbola, and from physical initial data it never reverses the time velocity at any coupling strength.

Three claim types appear below and are kept separate. Statements about the source’s published algebra are replication results, verified to machine precision. The identification of the midpoint closure with the Cayley

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transform, the conservation law of the continuous closure, and the component structure of the two maps are proved. Statements about which closure the full field theory prefers are open questions, and we make no claim about physical pair production anywhere in this paper.

II. THE REPLICATED CONFIGURATION

We follow the notation of Ref. [4] and use its equation numbers. An event with coordinates $x^\mu(\tau)$ evolves under the Lorentz force of five τ -dependent gauge fields (source Eq. 12), and may exchange mass with the field (source Eq. 14), which is what permits $\dot{t} = dt/d\tau$ to change sign. The incoming trajectory has velocity $u = \dot{t}_{\text{in}}(1, v, 0, 0)$ with $\dot{t}_{\text{in}} = 1/\sqrt{1-v^2}$ (source Eqs. 47–48). The nucleus induces the potential $a^0 = a^5 = (Ze/4\pi R)\delta(\tau - \tau_1)$ at interaction distance R in the small-correlation-length limit (source Eq. 56). The strength of the kick is the dimensionless coupling

$$g_e = \frac{\lambda}{M} \frac{Ze^2}{4\pi R^2}, \quad (1)$$

with λ the correlation length and M the Galilean target mass (source Eq. 64). Closing the impulse integrals with the midpoint velocities (source Eq. 63) yields a linear system (source Eq. 66) whose solution (source Eq. 67) has time component

$$\dot{t}_f = \frac{\dot{t}_{\text{in}} \left(1 - g_e v \hat{R}_x\right) + \frac{1}{4} g_e^2 (\dot{t}_{\text{in}} + 2)}{1 - \frac{1}{4} g_e^2}, \quad (2)$$

where $\hat{R}_x$ is the component of the unit vector to the interaction point along the incoming motion (source Eq. 76). The numerator is positive definite (source Eq. 78), so the sign of $\dot{t}_f$ is the sign of the denominator, giving reversal exactly for $g_e > 2$ (source Eqs. 79–80), and $\dot{t}_f \rightarrow -(\dot{t}_{\text{in}} + 2)$ as $g_e \rightarrow \infty$ (source Eq. 81).

Our transcription of the system and of the closed form agree with each other to 5.7×10^{-14} over a ninety-cell parameter sweep, the time component of the closed form matches Eq. (2) to 1.4×10^{-14} , the threshold is exact in every cell, and the asymptote is reproduced to 1.4×10^{-3} at $g_e = 1000$. The replication record, with a line-by-line provenance note mapping each source equation to code, is part of the repository described in Sec. VI.

III. THE REDUCED KICK SYSTEM AND ITS TWO CLOSURES

During the kick the source evaluates the potential at the fixed interaction point, which is what its delta collapse does in source Eq. 61. Writing w for the component of the spatial velocity along $\hat{R}$ and $s \in [0, 1]$ for the accumulated interaction kernel, the force equations (source Eqs. 59–60)

reduce to the linear system

$$\frac{d}{ds} \begin{pmatrix} \dot{t} + 1 \\ w \end{pmatrix} = G \begin{pmatrix} \dot{t} + 1 \\ w \end{pmatrix}, \quad G = -g_e \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (3)$$

with the perpendicular velocity component unchanged. The kernel-derivative part of the force is a total derivative whose contribution vanishes identically in the variable s . Everything below is a statement about the two ways of turning the generator G into a map from incoming to outgoing velocities.

Proposition 1 (Identification). *The midpoint closure of source Eq. 63 applied to the reduced system (3) is the Cayley transform $v_f = (I - G/2)^{-1}(I + G/2)v_{\text{in}}$, and its time component reproduces Eq. (2) identically.*

Proof. The midpoint convention evaluates the velocity in the accumulated kick as the average of the endpoint velocities, so $v_f - v_{\text{in}} = G(v_{\text{in}} + v_f)/2$, which rearranges to the stated transform. Expanding with $G^2 = g_e^2 I$ gives $v_f = [(1 + \frac{1}{4}g_e^2)I + G]v_{\text{in}}/(1 - \frac{1}{4}g_e^2)$, whose first component is Eq. (2) after substituting $w_{\text{in}} = v\dot{t}_{\text{in}}\hat{R}_x$. The identity was also verified numerically to 10^{-12} across the sweep of Sec. IV. $\square$

The Cayley transform is the Padé approximant of order $(1, 1)$ of the exponential map [6, 7]. Its eigenvalue action on the eigenvectors of G , which have eigenvalues $\mp g_e$, is $\lambda \mapsto (1 + \lambda/2)/(1 - \lambda/2)$, so one image eigenvalue is $(1 + g_e/2)/(1 - g_e/2)$. The pole at $g_e = 2$ is the published threshold.

Proposition 2 (Conservation and no reversal). *The continuous flow $\exp(sG)$ conserves $Q = (\dot{t} + 1)^2 - w^2$. For physical incoming data, meaning $w_{\text{in}} = v\dot{t}_{\text{in}}\hat{R}_x$ with $|\hat{R}_x| \leq 1$ and $|v| < 1$, the invariant is bounded below,*

$$Q \geq 2(\dot{t}_{\text{in}} + 1) \geq 4, \quad (4)$$

and the continuously accumulated kick satisfies $\dot{t}(s) + 1 \geq \sqrt{Q} \geq 2$ for all s and all g_e . Hence $\dot{t}(s) \geq 1$ throughout. The time velocity therefore not only never reverses, it never falls below its rest value.

Proof. G satisfies $G^T \eta + \eta G = 0$ for $\eta = \text{diag}(1, -1)$, so $\exp(sG)$ lies in the invariance group of Q . In light-cone coordinates $(\dot{t} + 1) \pm w$ the flow is diagonal with factors $e^{\mp g_e s}$, so their product Q is constant and both coordinates remain positive if they start positive. Incoming data has $\dot{t}_{\text{in}} + 1 > \dot{t}_{\text{in}} > |w_{\text{in}}|$ whenever $|v| < 1$, since $|w_{\text{in}}| \leq v\dot{t}_{\text{in}} < \dot{t}_{\text{in}}$, so both light-cone coordinates start positive and stay positive.

The bound (4) is algebraic. Using $|\hat{R}_x| \leq 1$ and the identity $v^2 \dot{t}_{\text{in}}^2 = \dot{t}_{\text{in}}^2 - 1$,

$$Q = (\dot{t}_{\text{in}} + 1)^2 - w_{\text{in}}^2 \geq (\dot{t}_{\text{in}} + 1)^2 - (\dot{t}_{\text{in}}^2 - 1) = 2(\dot{t}_{\text{in}} + 1),$$

and $\dot{t}_{\text{in}} \geq 1$ gives $Q \geq 4$. The first inequality is saturated when the interaction point lies along the incoming motion,

and the second as the event approaches rest. Taking the positive root, $\dot{t}(s) + 1 \geq \sqrt{Q} \geq 2$. The closed form $\dot{t}(s) + 1 = (\dot{t}_{\text{in}} + 1) \cosh(g_e s) - w_{\text{in}} \sinh(g_e s)$ was verified against an independent fixed-step integration to 10^{-9} . Over the ninety-cell sweep of Sec. IV the smallest invariant is $Q = 4.0445$, attained at the smallest sampled velocity with the interaction point most nearly along the motion, where the bound (4) reads 4.0412. $\square$

Proposition 3 (Component structure). *The invariance group of Q has disconnected components. For $g_e < 2$ the Cayley transform lies in the identity component and maps the physical branch $\dot{t} + 1 \geq \sqrt{Q}$ to itself. For $g_e > 2$ both of its eigenvalues are negative, so it is the product of a boost with the total reflection $-I$, lands in the component that reverses both $\dot{t} + 1$ and w , and maps the physical branch to the opposite branch $\dot{t} + 1 \leq -\sqrt{Q}$. No continuous one-parameter flow of G reaches that component.*

Proof. The image eigenvalues $(1 \mp g_e/2)/(1 \pm g_e/2)$ are both positive for $g_e < 2$ and both negative for $g_e > 2$, with product one in both cases, so the map is a Q -preserving matrix equal to a boost for $g_e < 2$ and to $-I$ times a boost for $g_e > 2$. The exponential of a fixed generator is connected to the identity, so it cannot equal a map with negative eigenvalues. Branch preservation and branch exchange follow from the sign of the eigenvalues on the light-cone coordinates. $\square$

The two closures are often described as competing conventions of equal standing. For this reduced system they are not, and the reason is that the generator is constant.

Proposition 4 (Regularization). *Let δ_ϵ be any smooth family of unit-mass kernels converging to the impulse. Driving the reduced system (3) with δ_ϵ gives the exact solution map $\exp(G)$ for every ϵ , independent of the width and of the shape of the kernel. The continuous closure is therefore the common limit of every smooth regularization of the impulsive potential, and for $g_e > 2$ the midpoint closure is the limit of none of them.*

Proof. Driving the system with the kernel gives $dV/d\tau = \delta_\epsilon(\tau)GV$ with G the constant matrix of Eq. (3). The generator at one parameter value commutes with the generator at every other, since both are multiples of the same fixed matrix, so the time-ordered exponential collapses to an ordinary one and

$$V(+\infty) = \exp\left(G \int \delta_\epsilon(\tau) d\tau\right) V(-\infty) = \exp(G) V(-\infty)$$

by unit mass. The result contains no reference to ϵ or to the profile, so the limit $\epsilon \rightarrow 0$ is $\exp(G)$ exactly rather than in the sense of an approximation. By Proposition 3 the midpoint map has two negative eigenvalues for $g_e > 2$ while $\exp(G)$ has two positive ones at every coupling, so beyond the pole the midpoint map is not the limit of any such family. $\square$

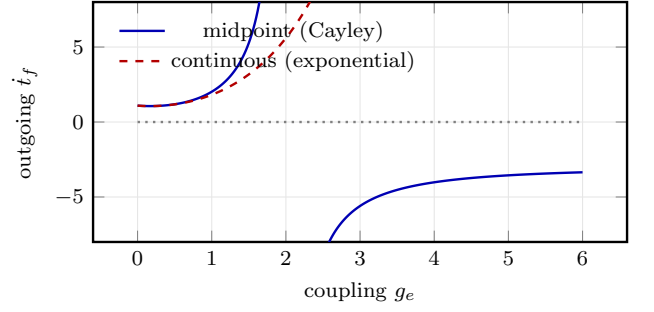


FIG. 1. Outgoing time velocity against coupling for the reference scattering geometry. The midpoint closure diverges at the pole $g_e = 2$ and emerges reversed, which is the published pair event. The continuous closure of the same generator grows monotonically and never reverses. Both curves are computed from the committed replication code.

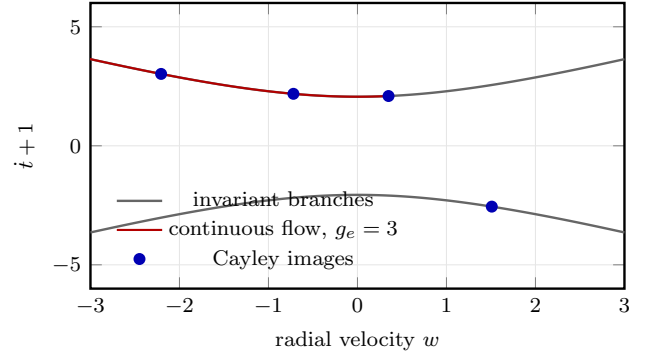


FIG. 2. The conserved-form picture. Physical initial data sits on the upper branch of the invariant hyperbola. The continuous flow moves along that branch and cannot leave it. The Cayley images move along the upper branch as the coupling grows toward two, then reappear on the lower branch beyond the pole. The Cayley transform of an $\mathfrak{so}(1, 1)$ element preserves the invariant exactly, so the residual here measures arithmetic and nothing else. Every plotted image satisfies the invariant to 4.1×10^{-13} , asserted at 10^{-12} in the committed data script. The residual is at most 6×10^{-15} at couplings away from the pole and reaches the quoted size only at $g_e = 1.9$ and $g_e = 2.1$, where the factor $(1 - g_e^2/4)^{-1}$ costs about three digits to cancellation.

The scope of Proposition 4 is exactly the scope of the reduction. It uses only that the kick generator is a fixed matrix during the interaction, which is what the frozen-point delta collapse supplies. In a setting where the generator varies with the event's own position or velocity the commutator no longer vanishes, the time-ordering does not collapse, and the argument gives nothing. That boundary is where a reader should press, and Sec. V states what remains open on the other side of it.

IV. NUMERICAL VERIFICATION

All numbers are from the committed evidence record of the replication run. The transcription guards give agreement between the printed linear system and the printed closed form at 5.7×10^{-14} and between the closed form and its time component at 1.4×10^{-14} . The continuous closed form of Proposition 2 agrees with an independent Runge–Kutta integration to better than 10^{-9} at couplings one, three, and six. At the reference geometry with $v = 0.4$ and $\hat{R}_x = 0.8$ the two closures give outgoing time velocities of 2.02 against 1.82 at coupling one, -5.60 against $+16.55$ at coupling three, and -3.35 against $+350.4$ at coupling six, the midpoint values approaching the printed asymptote from below while the continuous values grow exponentially.

One further reading of the sweep deserves to be stated on its own rather than listed among the guards. Exactly twenty-seven of the ninety cells, thirty percent, have spacelike outgoing velocities at intermediate coupling. This is neither a numerical failure nor a defect of the source. A worldline whose time velocity changes sign must pass through $\dot{t} = 0$, and there the outgoing velocity cannot be timelike, so any mechanism that produces reversal by turning a coupling continuously must produce spacelike outgoing data on the way through. The spacelike window is part of the reversal mechanism, not an artifact beside it. The source’s timelike statement is a large-coupling statement and is consistent with this. The practical edge is for campaigns that count reversals, since a filter that discards spacelike outgoing states will also discard part of the very effect being counted.

V. WHAT IS AND IS NOT IMPLIED

The published calculation is internally exact and our replication confirms every printed consequence of its stated conventions. The midpoint convention is, moreover, not a numerical shortcut that the source could simply have avoided. It is the closure that defines the impulsive model, adopted for the same reason implicit midpoint methods are prized in geometric integration, and the resulting map does preserve the invariant form of the reduced system exactly [6]. The point established here is narrower and, we think, sharper. Within the impulsive model the pair threshold is located exactly at the pole where the Cayley approximant of the kick exponential leaves the identity component of the invariance group, and the continuously accumulated version of the same kick provably never produces a reversal. The classical pair event of this configuration is therefore a property of the closure prescription, not of the accumulated force.

Within the reduced system we can say more than that the closure decides. Proposition 4 shows that the continuous closure is not one convention among two but the exact solution for every smooth kernel of unit mass, at every width, so it is the common limit of all of them. The midpoint map beyond the pole is not that limit for any of them, because it sits in the wrong component. Inside the reduction, then, the question of which closure represents an impulsive but physical field is not open, and the continuous one wins it on uniqueness rather than on taste.

What stays open is whether the reduction survives. The argument uses only that the generator is a fixed matrix during the interaction, which is what the frozen-point delta collapse supplies and what the full theory does not. There the potential evolves with the moving event, radiation is present, and the interaction window is determined self-consistently, so the generator varies along the trajectory, the commutator that made the time-ordering collapse no longer vanishes, and Proposition 4 says nothing. A configuration in which the full pre-Maxwell field dynamics produces or forbids a reversal threshold would decide the physical question, and we state it as such rather than prejudice it.

There is also a practical corollary for any simulation campaign that counts time reversals, including ours. A discretization prescription can manufacture a reversal threshold at the breakdown radius of its own approximant, complete with a physically appealing energy condition, while the continuous dynamics it approximates has none. Reversal statistics inherit the component structure of the map used to close each interaction, so a campaign that measures them must either integrate continuously through interactions or preregister its closure prescription and test sensitivity to it. This lesson is now part of the anti-circularity contract of the pair-production campaign that produced this replication.

VI. METHODS AND REPRODUCIBILITY

The replication lives in a repository whose instruments were validated under sealed tolerance freezes before any result-bearing run, with every numerical bar fixed in advance. The transcription is guarded by solving the printed linear system numerically against the printed closed form across a parameter sweep, and the provenance note maps each source equation to its implementation line by line. Evidence records are append-only JSON with SHA-256 hashes of inputs and outputs, and every figure in this paper is generated by a committed script whose inputs are the committed evidence records or the committed replication code, with the invariant of Fig. 2 asserted inside the data script itself.

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